전산화학 (2011년 1학기)

3차 시험 해답

(**총 9 문제, 130점**)

1. **(총 10 점, 각 5 점)**

(a) 비교판정법을 이용하면

$$\frac{\ln 1}{1}+\frac{\ln 2}{2}+\frac{\ln 3}{3}+\frac{\ln 4}{4}+\frac{\ln 5}{5}+\cdots \;>\; \frac{\ln 2}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\cdots$$
$$=\left(\frac{\ln 2}{2}-\frac{1}{1}-\frac{1}{2}\right)+\left(\frac{1}{1}+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\cdots\right)=\infty$$

발산한다.

(b) 비교판정빕을 이용하면

$$\sum_{n=1}^{\infty}\frac{2^n}{3^n+n}=\frac{2}{3+1}+\frac{2^2}{3^2+2}+\frac{2^3}{3^3+3}+\cdots<\frac{2}{3}+\frac{2^2}{3^2}+\frac{2^3}{3^3}+\cdots=\frac{\frac{2}{3}}{1-\frac{2}{3}}=2$$

수렴한다.

2. **(총 10 점, 각 5 점)**

(a) $a_n = x^n/(2n-1)$ 이므로,

$$r=\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| \;=\; \lim_{n\to\infty}\left|\frac{x^{n+1}/(2n+1)}{x^n/(2n-1)}\right|$$
$$=\lim_{n\to\infty}\left|x\frac{2n-1}{2n+1}\right|=|x|<1\Rightarrow -1<x<1.$$

양 끝점에서 수렴 여부를 판단하기 위하여 다른 판정법을 이용한다.

먼저 $x=-1$ 에서

$$-1+\frac{1}{3}-\frac{1}{5}+\frac{1}{7}-\cdots=-\left(1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\cdots\right)$$

교대급수 판정법을 이용하면 위 교대급수는 수렴한다.

먼저 $x=1$ 에서

$$1+\frac{1}{3}+\frac{1}{5}+\frac{1}{7}+\cdots>\frac{1}{2}+\frac{1}{4}+\frac{1}{6}+\frac{1}{8}+\cdots=\frac{1}{2}\left(1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\cdots\right)\to\infty$$

$x=1$ 에서 발산한다.

그러므로 수렴구간은 $-1\le x<1$ 이다.

(b) $a_n = x^{2n+1}/(2n+1)!$ 이므로,

$$\begin{aligned} r = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| &= \lim_{n\to\infty}\left|\frac{x^{2n+3}/(2n+3)!}{x^{2n+1}/(2n+1)!}\right| = \lim_{n\to\infty}|\frac{x^2}{(2n+3)(2n+2)}| < 1 \\ &\Rightarrow \quad |x| < \lim_{n\to\infty}(2n+3)(2n+2) = \infty \\ &\Rightarrow \quad -\infty < x < \infty. \end{aligned}$$

3. **(총 20 점, 각 5 점)**

(a) $t^3/(1+t^2)$ 은 기함수이고 적분 범위가 $-2 \le t \le 2$ 이므로 적분값은 0 이다.

(b)

$$\begin{aligned} \int_0^a dx \int_0^{x^3} dy e^{y/x} &= \int_0^a dx \Big[x e^{y/x} \Big]_0^{x^3} \\ &= \int_0^a dx (x e^{x^2} - x) \\ &= \left[\frac{1}{2} e^{x^2} - \frac{1}{2} x^2 \right]_0^a \\ &= \frac{1}{2}\Big(e^{a^2} - a^2 - 1 \Big) \end{aligned}$$

(c) $x(x+1) = t$ 라 치환하면 $(2x+1)dx = dt$ 가 된다.

$$\begin{aligned} \int_0^{\pi/2} \sin\big[x(x+1)\big](2x+1)dx &= \int_0^{\pi/2(\pi/2+1)} \sin t dt \\ &= \Big[-\cos t \Big]_0^{\pi/2(\pi/2+1)} \\ &= -\cos\big(\pi/2(\pi/2+1)\big) + 1 \end{aligned}$$

(d) 부분적분법 이용

$u = v^2$ 그리고 $w' = v e^{-mv^2/2kT} dv$, 그러면 $u' = 2v dv$, $w = -\frac{kT}{m} e^{-mv^2/2kT}$.

$$\begin{aligned} \int_0^\infty v^3 e^{-mv^2/2kT} dv &= \left[-\frac{kT}{m} v^2 e^{-mv^2/2kT} \right]_0^\infty + \frac{kT}{m} \int_0^\infty 2v e^{-mv^2/2kT} dv, \\ &= -2\left(\frac{kT}{m}\right)^2 \left[e^{-mv^2/2kT} \right]_0^\infty = 2\left(\frac{kT}{m}\right)^2. \end{aligned}$$

4. **(10 점)**

$$
\begin{aligned}
w &= -\int_{\bar{V}_1}^{\bar{V}_2} P d\bar{V} \\
&= -\int_{\bar{V}_1}^{\bar{V}_2} \left(\frac{RT}{\bar{V}-B} - \frac{A}{T^{1/2}\bar{V}(\bar{V}+B)} \right) d\bar{V} \\
&= -\int_{\bar{V}_1}^{\bar{V}_2} \left\{ \frac{RT}{\bar{V}-B} - \frac{A}{BT^{1/2}} \Big(\frac{1}{\bar{V}} - \frac{1}{\bar{V}+B} \Big) \right\} d\bar{V} \\
&= -\left[RT\ln(\bar{V}-B) - \frac{A}{BT^{1/2}} \Big(\ln\bar{V} - \ln(\bar{V}+B) \Big) \right]_{\bar{V}_1}^{\bar{V}_2} \\
&= -\left\{ RT\ln(\bar{V}_2-B) + \frac{A}{BT^{1/2}} \Big(\ln\bar{V}_2 - \ln(\bar{V}_2+B) \Big) \right\} + \left\{ RT\ln(\bar{V}_1-B) - \frac{A}{BT^{1/2}} \Big(\ln\bar{V}_1 - \ln(\bar{V}_1+B) \Big) \right\} \\
&= -RT\ln\left(\frac{\bar{V}_2-B}{\bar{V}_1-B} \right) - \frac{A}{BT^{1/2}} \ln\left(\frac{\bar{V}_1(\bar{V}_2+B)}{\bar{V}_2(\bar{V}_1+B)} \right)
\end{aligned}
$$

5. **(10점)**

$$\frac{1}{2}\ln\left(\frac{1+x}{1-x}\right) = \frac{1}{2}\big(\ln(1+x) - \ln(1-x)\big).$$

한편,

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

그리고

$$\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

이므로,

$$
\begin{aligned}
\frac{1}{2}\ln\left(\frac{1+x}{1-x}\right) &= \frac{1}{2}\left(2x + 2\frac{x^3}{3} + 2\frac{x^5}{5} + \cdots \right) \\
&= \left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots \right) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{2n+1}
\end{aligned}
$$

6. **(총 20점, 각 5점)**

(a)

$$\int_C (xy^2 dx + x^2 y dy),$$

위 식에서 경로 C는 점 $(1,1)$부터 점 $(5,5)$을 연결하는 직선 $y-x=0$을 의미한다.

$$
\begin{aligned}
\int_C (y\cos x dx + \sin x dy) &= \int_1^5 x(x)^2 dx + \int_1^5 (y)^2 y dy \\
&= \left[\frac{1}{4}x^4 \right]_1^5 + \left[\frac{1}{4}y^4 \right]_1^5 \\
&= \frac{1}{4}\big(5^4 - 1\big) + \frac{1}{4}\big(5^4 - 1\big) \\
&= 312
\end{aligned}
$$

(b)

$$\int_C (xy^2dx + x^2ydy),$$

위 식에서 경로 C는 처음에는 $(1,1)$에서 $(5,1)$까지 그 다음에는 $(5,1)$에서 $(5,5)$로 주어진다.

$$\begin{aligned}\int_C (xy^2dx + x^2ydy) &= \int_1^5 x(1)^2dx + \int_1^5 (5)^2ydy \\ &= \left[\frac{1}{2}x^2\right]_1^5 + 25\left[\frac{1}{2}y^2\right]_1^5 \\ &= \frac{1}{2}(25-1) + \frac{25}{2}(25-1) = 12 + 300 = 312\end{aligned}$$

(c) $M(x,y) = xy^2,\ N(x,y) = x^2y$ 0이므로

$$\left(\frac{\partial M}{\partial y}\right)_x = 2xy, \qquad \left(\frac{\partial N}{\partial x}\right)_y = 2xy \Rightarrow \left(\frac{\partial M}{\partial y}\right)_x = \left(\frac{\partial N}{\partial x}\right)_y$$

0이므로, $xy^2dx + x^2ydy$는 완전미분0이다.

(d) $C : (1,1) \to (5,5)$

$$\int_C (xy^2dx + x^2ydy) = \left[\frac{1}{2}x^2y^2\right]_{(1,1)}^{(5,5)} = \frac{1}{2}\big((5)^2(5)^2 - (1)^2(1)^2\big) = 312$$

7. **(10점)**

$$u(r) = \begin{cases} \infty & (0 < r < \sigma) \\ -\epsilon & (\sigma < r < \lambda\sigma) \\ 0 & (r > \lambda\sigma) \end{cases}$$

0이므로,

$$\begin{aligned}B(T) &= -2\pi \int_0^\infty [e^{-u(r)/(k_BT)} - 1]r^2dr \\ &= -2\pi \int_0^\sigma [e^{-u(r)/(k_BT)} - 1]r^2dr - 2\pi \int_\sigma^{\lambda\sigma} [e^{-u(r)/(k_BT)} - 1]r^2dr - 2\pi \int_{\lambda\sigma}^\infty [e^{-u(r)/(k_BT)} - 1]r^2dr \\ &= 2\pi \int_0^\sigma r^2dr - 2\pi \int_\sigma^{\lambda\sigma} [e^{\epsilon/(k_BT)} - 1]r^2dr \\ &= 2\pi \left[\frac{1}{3}r^3\right]_0^\sigma - 2\pi(e^{\epsilon/(k_BT)} - 1)\left[\frac{1}{3}r^3\right]_\sigma^{\lambda\sigma} \\ &= \frac{2\pi\sigma^3}{3} - \frac{2\pi\sigma^3}{3}(\lambda^3 - 1)(e^{\epsilon/(k_BT)} - 1) \\ &= \frac{2\pi\sigma^3}{3}\big[1 - (\lambda^3 - 1)(e^{\epsilon/(k_BT)} - 1)\big]\end{aligned} \tag{1}$$

8. **(총 20점)**

(a) **(10 점)**

$$f(x) = L^2 - x^2 \tag{2}$$

$f(x)$ 가 우함수이므로 Fourier 급수의 계수 a_n 은 다음과 같이 구할 수 있다.

$$\begin{aligned}
a_n &= \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx, \\
&= \frac{2}{L}\int_{0}^{L} (L^2 - x^2)\cos\left(\frac{n\pi x}{L}\right)dx, \\
&= 2L\int_{0}^{L} \cos\left(\frac{n\pi x}{L}\right)dx - \frac{2}{L}\int_{0}^{L} x^2\cos\left(\frac{n\pi x}{L}\right)dx, \\
&= 2L\left[\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\right]_0^L - \frac{2}{L}\left[\frac{L}{n\pi}x^2\sin\left(\frac{n\pi x}{L}\right)\right]_0^L \\
&\quad + \frac{2}{L}\int_{0}^{L} \frac{2L}{n\pi}x\sin\left(\frac{n\pi}{L}x\right)\Bigg]_0^L \quad (\text{부분적분 } 0\text{이용}) \\
&= \frac{4}{n\pi}\int_{0}^{L} x\sin\left(\frac{n\pi x}{L}\right)dx \qquad (3) \\
&= \frac{4}{n\pi}\left[-\frac{L}{n\pi}x\cos\left(\frac{n\pi x}{L}\right)\right]_0^L + \frac{4}{n\pi}\int_{0}^{L}\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)dx \\
&= -\frac{4L^2}{(n\pi)^2}\cos(n\pi) + \frac{4L}{(n\pi)^2}\left[\sin\left(\frac{n\pi x}{L}\right)\right]_0^L \\
&= -\frac{4L^2}{(n\pi)^2}(-1)^n \\
&= \frac{4L^2}{(n\pi)^2}(-1)^{n+1} \qquad (4)
\end{aligned}$$

한편, $f(x)$ 가 우함수이므로 a_0 는

$$\begin{aligned}
a_0 &= \frac{1}{L}\int_{-L}^{L} f(x)dx \\
&= \frac{2}{L}\int_{0}^{L} (L^2 - x^2)dx \\
&= \frac{2}{L}\left[L^2x - \frac{1}{3}x^3\right]_0^L dx \\
&= \frac{4}{3}L^2 \qquad (5)
\end{aligned}$$

Fourier 급수의 계수 b_n 은 다음과 같이 주어지고

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx,$$

$f(x)$ 가 우함수이고 $\sin(n\pi x/L)$ 은 기함수이므로 $b_n = 0$ 이 된다.

(4) 식과 (5) 식으로부터 Fourier 급수는 다음과 같이 구해진다.

$$f(x) = \frac{2L^2}{3} + \sum_{n=1}^{\infty} \frac{4L^2}{(n\pi)^2}(-1)^{n+1} \cos\left(\frac{n\pi x}{L}\right) \tag{6}$$

(b) **모두 맞게 처리**

(c) (6) 식에 $x = 0$ 을 대입하면

$$f(0) = \frac{2L^2}{3} + \sum_{n=1}^{\infty} \frac{4L^2}{(n\pi)^2}(-1)^{n+1} \tag{7}$$

위 식을 전개하면

$$= \frac{2L^2}{3} + \frac{4L^2}{\pi^2}\left(\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots\right)$$

한편, (2) 식으로부터 $f(0) = L^2$ 이므로, 정리하면

$$L^2 = \frac{2L^2}{3} + \frac{4L^2}{\pi^2}\left(\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots\right)$$

$$\Rightarrow \boxed{\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots = \frac{\pi^2}{12}} \tag{8}$$

9. **(총 20 점)**

(a) **(5 점)** Because

$$P = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2} \tag{9}$$

이므로

$$\boxed{\left(\frac{\partial P}{\partial V}\right)_T = -\frac{nRT}{(V - nb)^2} + \frac{2n^2 a}{V^3}}$$

(b) **(5 점)**

$$\boxed{\left(\frac{\partial^2 P}{\partial V^2}\right)_T = \frac{\partial}{\partial V}\left(\frac{\partial P}{\partial V}\right)_T = \frac{\partial}{\partial V}\left[-\frac{nRT}{(V - nb)^2} + \frac{2n^2 a}{V^3}\right] = \frac{2nRT}{(V - nb)^3} - \frac{6n^2 a}{V^4}}$$

(c) **(10 점)**

$$\left(\frac{\partial P}{\partial V}\right)_T = -\frac{nRT_c}{(V_c - nb)^2} + \frac{2n^2 a}{V_c^3} = 0, \tag{10}$$

$$\left(\frac{\partial^2 P}{\partial V^2}\right)_T = \frac{2nRT_c}{(V_c - nb)^3} - \frac{6n^2 a}{V_c^4} = 0, \tag{11}$$

0이므로 식 (10)과 식 (11)를 정리하면,

$$\frac{nRT_c}{(V_c - nb)^2} = \frac{2n^2a}{V_c^3}, \tag{12}$$

$$\frac{2nRT_c}{(V_c - nb)^3} = \frac{6n^2a}{V_c^4}. \tag{13}$$

식 (12)과 식 (13)로부터

$$\frac{V_c - nb}{2} = \frac{V_c}{3} \Rightarrow \boxed{V_c = 3nb} \tag{14}$$

식 (14)을 식 (12)에 대입하면

$$\frac{nRT_c}{(3nb - nb)^2} = \frac{2n^2a}{(3nb)^3} \Rightarrow \boxed{T_c = \frac{8a}{27Rb}} \tag{15}$$

식 (14)과 식 (15)을 식 (9)에 대입하면

$$\boxed{P_c = \frac{nR(8a/27Rb)}{3nb - nb} - \frac{an^2}{(3nb)^2} = \frac{a}{27b^2}} \tag{16}$$